

Mega-Constellation Design for Integrated Satellite-Terrestrial Networks for Global Seamless Connectivity

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Abstract—In this letter, we study the integrated satellite-terrestrial network where users can access the core network by the backhaul links via both multi-layer satellites in the space and terrestrial access points on the ground. The mega-constellation design for such an integrated satellite-terrestrial network is investigated to realize the global seamless connectivity and provide high-rate backhaul transmission. First, we propose a theoretical framework for the average uplink capacity analysis. Second, based on the developed theoretical framework, we design a mega satellite constellation given the capacity of terrestrial backhaul links for realizing the global connectivity with the minimum satellite number. Simulation results show that as the terrestrial infrastructures are distributed more evenly in latitude, the designed mega-constellation first requires more satellites and then remains unchanged regardless of the terrestrial transmission capacity of high-latitude user terminals.

Index Terms—Integrated satellite-terrestrial network, mega constellation, satellite number minimization.

I. INTRODUCTION

WITH the development of 5G mobile network, various mobile applications, e.g., automotive driving or smart city, have triggered the growing needs for the ubiquitous coverage and high-rate transmission services [1]. Future 6G mobile network is expected to provide the continuous network connection at anywhere and anytime [2]. However, the terrestrial infrastructures are not sufficient to support the full-coverage high-rate backhaul services due to the limited resources and the knotty deployment of access points [3]. Benefitted from the high operating frequency and vast coverage, the satellite communication network is a promising solution for the seamless user access and high-rate backhaul services around the world [4]. Making use of both satellite and terrestrial fiber links, user terminals (UTs) access the core network [5].

Most current satellite constellations are independently constituted by single-layer low Earth orbit (LEO) satellites [6], [7] to provide global backhaul transmission service. Specifically, a global LEO constellation is designed in [7] providing the quality-of-service-guaranteed global backhaul transmission service. However, given the limited orbit resources on a single satellite layer, multi-layer satellite constellation is widely applied in practice, e.g., Starlink and Telesat, and thus, the multi-layer constellation design needs to be considered.

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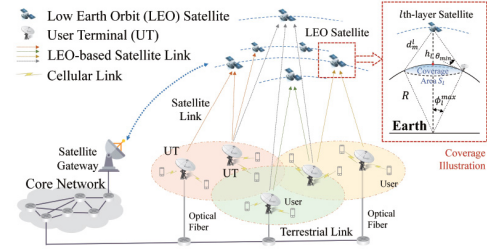


Fig. 1. Integrated satellite-terrestrial network model.

Moreover, the influence of the terrestrial network on constellation design is ignored in current works. In practice, the satellite network is constructed on top of the terrestrial network, and thus, the satellite deployment relies heavily on the distribution and terrestrial transmission capacity of user terminals. Therefore, the satellite and terrestrial networks need to be considered jointly for the satellite constellation design.

To handle these issues, we discuss the mega-constellation design for the integrated satellite-terrestrial network given the terrestrial terminal deployment, aiming to support both global seamless connectivity and high-rate backhaul transmission. Against this background, two questions remain to be answered:

- First, how to depict the uplink capacity of the integrated backhaul network influenced by the number of satellites and the terrestrial transmission capacity of UTs?
- Second, how to design the mega-constellation for multi-layer satellites to fulfill the rate-differentiated transmission demands with the minimum number of satellites?

To address the aforementioned challenges, we propose an uplink capacity theoretical framework where the average uplink capacity is analyzed given the terrestrial distribution density of UTs. Based on the data rate of satellites analyzed in the theoretical framework, we optimize the mega-constellation of multi-layer satellites to minimize the total required satellites number.

II. SYSTEM MODEL

In this section, we first describe the network structure, and then introduce the satellite coverage and transmission model.

A. Network Structure

As illustrated in Fig. 1, we consider an integrated satellite-terrestrial network consisting of the UTs as well as multiple layers of LEO satellites on different altitudes. The terrestrial users transmit the data to the UTs over the C-band spectrum. The UTs then access the core network via both two types of backhaul links: through the *terrestrial link* once the UT is within the range of the optical network and through the *satellite link* via either layer of LEO satellite¹ at anywhere.

¹Assume that the UTs are equipped with the GNSS receivers, being able to estimate the position of the satellite, and thus, the Doppler shift of LEO satellites can be significantly compensated [2], [8].

Specifically, the satellite links are established over the Ka-band spectrum and all satellites deliver the data to the core network via the satellite gateways.

B. Satellite Coverage Model

We discuss the coverage of the satellite on layer l assuming that the Earth is a sphere with radius R . The coverage area of a satellite is determined by its altitude h_l and the elevation angle range of the line-of-sight (LoS) terrestrial-satellite link. Let θ_{min} represent the minimum elevation angle of the UT, so the UT is considered within the coverage of a satellite when its elevation angle $\theta_l \geq \theta_{min}$. Therefore, the coverage area S_l of a satellite is the area of a spherical crown with the angular radius $\phi_l^{max} = \arccos(\frac{R}{R+h_l} \cos \theta_{min}) - \theta_{min}$, expressed by

$$S_l = 2\pi R^2(1 - \cos \phi_l^{max}). \quad (1)$$

C. Satellite Transmission Model

We discuss the data transmission of the UT to the satellite within time slot² t . For the brevity of expression, the index t is omitted. The antenna arrays are assumed at both satellites and UTs for the directional beamforming. For the tractability of the analysis, an approximated sectorized array model is widely adopted for formulating the antenna gain of each beam [9]. Single-beam is discussed for the satellite that all the UTs connected to that satellite are within its main lobe with the gain G_{sate}^M . Multi-beam is assumed for each UT to connect with multiple satellites, where the antenna gain of UT's each beam is G_{ut}^M for the desired link in the main lobe and G_{ut}^S for the interference link in the side lobe. Considering the satellite link from UT q to satellite m on the l -th layer, the signal to interference plus noise ratio (SINR) $\gamma_{m,q}^l$ at the satellite is

$$\gamma_{m,q}^l = \frac{PG_{ut}^M G_{sate}^M h_{r,qm,p} (d_{m,q}^l)^{-\alpha}}{(\sum_{n=1}^L k_n)(\sigma^2 + I_{m,q}^l)}. \quad (2)$$

where P is the UT's total transmitting power and k_n denotes the average number of connected satellites on the n -th layer. An equal power scheme is used by the UT to allocate the same transmitting power to each satellite backhaul link. The rain fading $h_{r,qm,p} = 10^{-\frac{A_{r,qm,p}}{10}}$ is considered over the Ka band, where $A_{r,qm,p}$ [dB] denotes the rain attenuation exceeded for $p\%$ of an average year, which is determined by the geometrical and electrical characteristics of the antenna [10]. For convenience, we define $G_0 = G_{sate}^M G_{ut}^M h_r$, where h_r is the expectation of the rain fading. $d_{m,q}^l$ denotes the distance between UT q and satellite m on the l -th layer, and α is the path loss exponent. The additive white gaussian noise is considered following $\mathcal{CN}(0, \sigma^2)$ distribution, and $I_{m,q}^l$ is the interference.

We assume that all satellites share the same spectrum resource pool with total bandwidth B for the data transmission. In total J orthogonal subchannels are divided and allocated by each satellite to the UTs within its coverage according to their satellite transmission rate requirements. Thus, the transmission rate $R_{m,q}^l$ of the satellite link from UT q to satellite m on the l -th layer is influenced by the number $V_{m,q}^l$ of subchannels allocated to UT q , which is expressed by

$$R_{m,q}^l = \frac{BV_{m,q}^l}{J} \log(1 + \gamma_{m,q}^l). \quad (3)$$

²The network topology is viewed as static in each time slot with 1s duration.

Note that any UT q can be served by multiple satellites simultaneously once the elevation angle of the satellite is not smaller than θ_{min} for UT q . Let $\mathcal{A}_l^q$ denote the set of serving satellites on layer l for UT q , where $\mathcal{A}_l^q = \{m \mid \theta_m^l \geq \theta_{min}\}$. Considering L satellite layers, the total data rate of UT q is

$$R_q = \sum_{l=1}^L \sum_{m \in \mathcal{A}_l^q} R_{m,q}^l. \quad (4)$$

III. UPLINK CAPACITY THEORETICAL FRAMEWORK

In this section, we propose an uplink capacity theoretical framework for the integrated satellite-terrestrial network.

A. Interference Model

Given the on-demand deployment of the UTs, their locations are commonly assumed to follow the homogeneous Poisson point process (HPPP) Φ with density λ UTs/km². The rate requirement of the UT's backhaul service is represented by C_{th} . The average coverage ratio of the terrestrial network is denoted by ρ . Each UT within the coverage of the terrestrial network is provided by the transmission capacity of C_{terr} .

The orthogonal subchannels are allocated by the satellite to the UTs randomly within its coverage, and the allocated number is proportional to the satellite rate requirements. Thus, the distribution of the interfering UT on the same subchannel for UT q follows the HPPP with density $\lambda' = \frac{\lambda}{J}$ UTs/km². The interference of UT q -satellite m (layer l) link is

$$I_{m,q}^l = V_{m,q}^l \sum_{n=1}^L \sum_{q' \in D_{m,q}^n} \frac{PG_{ut}^S G_{sate}^M h_{r,q'm,p}}{K} (d_{m,q'}^l)^{-\alpha}, \quad (5)$$

where $K = \sum_{n=1}^L k_n$ and $D_{m,q}^n$ is given in Proposition 1. Based on Campells theorem, the average interference of that satellite backhaul link $I_{m,q}^{l,(in)}$ (within the terrestrial coverage) and $I_{m,q}^{l,(out)}$ (outside the terrestrial coverage) are expressed by

$$\mathbb{E}[I_{m,q}^{l,(s)}] = \frac{PG_I}{K} \sum_{n=1}^L \frac{C_{sate}^{(s)} J \lambda'}{(C_{th} - C_{terr} \rho) \lambda S_l} \int_{q' \in D_{m,q}^n} (d_{m,q'}^l)^{-\alpha} dq'. \quad (6)$$

where $s \in \{in, out\}$ and $G_I = G_{ut}^S G_{sate}^M h_r$. We define the rate requirement of the satellite backhaul link by $C_{sate}^{(s)}$, where $C_{sate}^{(in)} = C_{th} - C_{terr}$ when $s = in$ and $C_{sate}^{(out)} = C_{th}$ when $s = out$. Given the orthogonal subchannel among different UTs, the average allocated subchannel number is proportional to the ratio of the rate requirement of UT q to the total rate requirement in the coverage area S_l of the same satellite.

Proposition 1: The interference region $D_{m,q}^n$ of co-layer and cross-layer interfering UTs is:

$$D_{m,q}^n = \begin{cases} \left\{ q' \mid d_o^l \leq d_{m,q'}^l \leq \sqrt{h_l^2 + 2Rh_l} \right\}, & n = l, \\ \left\{ q' \mid h_l \leq d_{m,q'}^n \leq \sqrt{h_l^2 + 2Rh_l} \right\}, & n \neq l. \end{cases}$$

The smallest distance d_o^l from the UT within coverage of the neighbouring satellite to satellite m satisfies that

$$(d_o^l)^2 = R^2 + (R + h_l)^2 - 2R(R + h_l) \cos \phi_l^o, \quad (7)$$

where $\phi_l^o = \max\{0, \sqrt{\frac{8\pi R^2(1-\cos\phi_l^{max})}{2\sqrt{3}k_l}} - R \cdot \phi_l^{max}\}/R$ and k_l is the connected satellite number on layer l for each UT. The longest distance of the LoS path between the interfering UT q' and satellite m on layer l is indicated by $\sqrt{h_l^2 + 2Rh_l}$.

Proof: See Appendix A. ■

Therefore, we can obtain the average interference in (6):

$$\mathbb{E}[I_{m,q}^{l,(s)}] = \frac{2\pi R P G_I C_{sat}^{(s)}}{(\alpha-2)(C_{th}-C_{terr})K S_l} \left[\eta_l \frac{(d_o^l)^{2-\alpha} - (2Rh_l + h_l^2)^{\frac{2-\alpha}{2}}}{R + h_l} + \left(\sum_{n \neq l} \eta_n \frac{(h_l)^{2-\alpha} - (2Rh_l + h_l^2)^{\frac{2-\alpha}{2}}}{R + h_l} \right) \right], \quad (8)$$

where $\eta_l = k_l/(\sum_{n=1}^L k_n)$ represents the proportion of the satellite link number to the l -th layer of satellite (denoted by k_l) in all the satellite links of the same UT.

B. Uplink Capacity Analysis

In this part, we first derive the average spectrum efficiency $\mu_l^{(s)}$ of the satellite link from the UT (where $s \in \{in, out\}$) to the satellite on the layer l , and then derive the average data rate $C_{UT}^{(s)}$ of the UTs considering all the terrestrial fiber links. The uplink capacity of the integrated satellite-terrestrial network is calculated considering all UTs.

Proposition 2: The average spectrum efficiency $\mu_l^{(s)}$ is

$$\mu_l^{(s)} = \frac{\mathcal{F}((d_m^l)^2, X, Y) - \mathcal{F}((h_l)^2, X, Y)}{2R(R + h_l)(1 - \cos\phi_l^{max}) \ln 2}, \quad (9)$$

where the longest distance between satellite m (on layer l) and its serving UT is $d_m^l = \sqrt{(R \sin \theta_{min})^2 + h_l^2 + 2Rh_l - R \sin \theta_{min}}$. The function is defined by $\mathcal{F}(v, X, Y) = v[Y_2 F_1(1, \frac{1}{Y}; 1 + \frac{1}{Y}; -Xv^Y) + \ln(1 + Xv^Y) - Y]$ and we set that $X = \frac{P G_0}{K(\sigma^2 + \mathbb{E}[I_{m,q}^{l,(s)}])}$, $Y = -\frac{\alpha}{2}$.

Proof: See Appendix B. ■

The average data rate of each satellite link $\mathbb{E}[R_{m,q}^{l,(s)}]$ from the UT to the satellite on the l -th layer, denoted by $C_l^{(s)}$, is the product of the available bandwidth for the UT and the average spectrum efficiency $\mu_l^{(s)}$. According to the properties of PPP, we express $C_l^{(s)} = \mathbb{E}[R_{m,q}^{l,(s)}]$ as:

$$C_l^{(s)} = \sum_{u_1=1}^{\infty} \sum_{u_2=1}^{\infty} \left[(\rho \lambda S_l)^{u_1} \frac{e^{-\rho \lambda S_l}}{u_1!} \right] \left[((1-\rho) \lambda S_l)^{u_2} \frac{e^{-(1-\rho) \lambda S_l}}{u_2!} \right] \cdot \frac{C_{sat}^{(s)} B \cdot \mu_l}{(C_{th} - C_{terr}) \rho u_1 + C_{th} (1 - \rho) u_2}. \quad (10)$$

Given that the UT connects with k_l satellites on the l -th layer, the average data rate of the UT within and outside the terrestrial network coverage are expressed by

$$C_{UT}^{(in)} = \sum_{l=1}^L k_l C_l^{(in)} + C_{terr}, \quad C_{UT}^{(out)} = \sum_{l=1}^L k_l C_l^{(out)}. \quad (11)$$

Thus, the uplink capacity C for the entire integrated satellite-terrestrial network can be modeled by the sum rate of both satellite and terrestrial fiber links of all UTs.

$$C = \rho \lambda S_E \cdot C_{UT}^{(in)} + (1 - \rho) \lambda S_E \cdot C_{UT}^{(out)}, \quad (12)$$

where S_E denotes the area of the whole earth. The uplink capacity is related with the average number of satellites

k_l ($1 \leq l \leq L$) serving the same UT as well as the coverage ratio ρ and the transmission capacity C_{terr} of the terrestrial network. Given the density λ and data rate requirement C_{th} of the UTs, the uplink capacity C can be calculated by solving the equations (8)-(12).

IV. MEGA-CONSTELLATION DESIGN

In this section, we first formulate the satellite number minimization problem, and then design the mega-constellation using the dynamic programming based method.

A. Satellite Number Minimization Problem Formulation

We aim to minimize the total number of deployed satellites in all layers given the terrestrial optical network while satisfying the rate requirements of all UTs for backhaul transmission. Let N_l represent the number of orbits on layer l (with altitude h_l) and let M_i^l denote the number of satellites in orbit i on layer l . Thus, the satellite number minimization problem is

$$\min_{h, N, M} \sum_{1 \leq l \leq L} \sum_{i \in N_l} M_i^{(l)}, \quad (13)$$

$$s.t. \quad C_{UT}^{(s)} \geq C_{th}, \forall s \in \{in, out\}, \forall UT, \forall t \quad (13a)$$

where $C_{UT}^{(s)}$ is the average data rate of the UT inside ($s = in$) and outside ($s = out$) of the terrestrial network coverage.

B. Mega-Constellations Design

In this part, we design the mega-constellation for the integrated satellite-terrestrial network via dynamic programming, aiming to minimize the number of total deployed satellites.

1) *One-Coverage Satellite Constellation Deployment:* We first consider a one-coverage satellites constellation deployment (i.e., $L = 1, k = 1$) with polar orbits [6] on the same satellite layer l . For the seamless coverage, the number of satellites deployed on each orbit is assumed to be same, expressed by $M^{(l)} = M_1^{(l)} = M_2^{(l)} = \dots = M_i^{(l)}$. The satellites moves in the same direction with a phase difference $\pi/M^{(l)}$ between two neighbouring orbits. Considering the hemisphere divided by one polar orbit, to satisfy the seamless coverage of the equator, the coverage angular radius ϕ_l^{max} and the number of orbits N satisfy the relation that

$$(N-1)\phi_l^{max} + (N+1)\Delta_l = \pi, \quad (14)$$

where $\Delta_l = \arccos(\cos\phi_l^{max} / \cos(\pi/M^{(l)}))$ according to the geometry relation. We can express the relation among θ_{min} , h_l , N , and satellites number $M^{(l)}$ on each orbit as

$$(N+1) \arccos \left[\cos \left[\arccos \left(\frac{R \cos \theta_{min}}{R + h_l} \right) - \theta_{min} \right] / \cos \left(\frac{\pi}{M^{(l)}} \right) \right] + (N-1) \left[\arccos \left(\frac{R \cos \theta_{min}}{R + h_l} \right) - \theta_{min} \right] = \pi. \quad (15)$$

Definition 1: A UT q is $\mathbf{k}$ -covered (where $\mathbf{k} = \{k_l\}, 1 \leq l \leq L$) indicates that UT q is covered by k_l satellites on layer l , where k_l represents the l -th coverage degree of UT q for the satellites on layer l . The region is $\mathbf{k}$ -covered if each UT q within this region is $\mathbf{k}$ -covered.

Based on the uplink capacity framework in Section III, the average data rate $C_{UT}^{(s)}(\mathbf{k})$ of the UT can be reached when the whole discussed region is $\mathbf{k}$ -covered. Therefore, to satisfy the rate requirement C_{th} for any UT in any position, we need to

Algorithm 1 Satellite Number Minimization Algorithm

Input: Maximum satellite layer L ; Satellite orbit number N ; Satellite number per orbit $M^{(l)}$ for each layer; Required average data rate $C_{th}^{(s)}$, $\forall s$.

Output: Optimal satellite altitude $\mathbf{h}$, optimal coverage degree $\mathbf{k}_f$.

- 1: Calculate the altitude h_l for each layer l given $M^{(l)}$ according to (15).
- 2: Calculate the average rate $C_l^{(s)}$ of the l -th layer satellite link given (10).
- 3: Initialize satellite number: $\chi(1, t) = \lceil t/C_l^{(s)} \rceil NM^{(1)}$, $1 \leq t \leq C_{th}^{(s)}$.
- 4: **for** Layer number $2 \leq l \leq L$ **do**
- 5: Calculate the minimum satellite number $\chi(l, C_{th}^{(s)})$ according to (17).
- 6: Update the optimal coverage degree $\mathbf{k}_{th} = \arg \min_{\mathbf{k}} \chi(l, C_{th}^{(s)})$.
- 7: $l \leftarrow l + 1$.

realize that the whole earth is seamlessly $\mathbf{k}_{th}$ -covered so that $C_{UT}^{(s)}(\mathbf{k}_{th}) \geq C_{th}$, $\forall s \in \{in, out\}$, $\forall UT$.

2) *Global Coverage for the Required Data Rate:* Considering L satellite layers and terrestrial capacity C_{terr} , for providing the average data rate C_{th} , the satellite minimization problem (13) can be decoupled from reformulated as below.

$$\min_{\mathbf{h}, \mathbf{k}} \sum_{l=1}^L k_l N \cdot M^{(l)} \quad (16)$$

$$s.t. \quad C_{UT}^{(s)} \geq C_{th}, 0 \leq k_l \leq \left\lceil \frac{C_{sate}^{(s)}}{C_l^{(s)}(h_l)} \right\rceil, \forall s, \forall UT, \quad (16a)$$

where k_l is the coverage degree of the l -th satellite layer. Problem (16) is a complete knapsack problem, where the volume of each object is $NM^{(l)}$ and the weight of each object is $C_l^{(s)}(h_l)$. We aim to minimize the total number of satellites (i.e., total weight of the knapsack) while providing C_{th} data rate to each UT in any position (i.e., filling the knapsack with size C_{th}). Given that the number of satellites $M^{(l)}$ on each orbit is a natural number, the possible value of satellite altitude h_l is discrete with each feasible $M^{(l)}$ according to (15), and the reachable rate $C_l^{(s)}(h_l)$ of each satellite link is also discrete.

To solve the complete knapsack problem (16), we then propose a dynamic programming based satellite number minimization (DP-SNM) algorithm. Let $\chi(l, c)$ denote the minimum satellite number when deploying the satellites from layer 1 to layer l (initialed by L), where t denotes the remaining data rate required by the UTs (initialed by $C_{sate}^{(s)}$). The recursive relation between the minimum satellite number can be expressed by the following equation where $0 \leq k_l C_l^{(s)} \leq c$.

$$\chi(l, c) = \min_{k_l} \left\{ \chi(l-1, c - k_l C_l^{(s)}) + k_l NM^{(l)}, \left\lceil \frac{C_{sate}^{(s)}}{C_l^{(s)}} \right\rceil NM^{(l)} \right\}. \quad (17)$$

The DP-SNM algorithm is described in Algorithm 1. First, given $M^{(l)}$ for layer l , we calculate the optimal altitude h_l for each layer l and derive the data rate $C_l^{(s)}$ provided by the l -th layer of satellites to each satellite. Second, we initialize the minimum satellite number $\chi(1, c)$ when deploying the satellites on only one layer. Third, we calculate the total satellite number with increasing l according to the dynamic programming based recurrence relation (17). Using the DP-SNM algorithm, we can obtain the optimal $\mathbf{k}_{th}$ that provides seamless C_{th} data rate for any UT.

The complexity of Algorithm 1 is then analyzed. Denote the maximum number of each UT's connected co-layer

TABLE I
PARAMETER SETTING

Parameters	Value
Total transmitting power of each UT P	2 W
Antenna gain of the UT G_{ut}^M, G_{ut}^S	31.2 dBi, -10 dBi
Antenna gain of the satellite G_{sate}^M, G_{sate}^S	37.1 dBi, -6.75 dBi
Main lobe beamwidth of the satellite and UT	$60^\circ, 2.5^\circ$
Bandwidth for satellite data transmission B	800 MHz
Number of subchannels J for each satellite	1000
Noise density for Ka-band transmission σ^2	-174 dBm/Hz
Density of UTs λ	4×10^{-6} km $^{-2}$
Minimum elevation angle θ_{min}	10°
Number of one-coverage satellite orbits N	6
Number of feasible satellites layers	40

Terrestrial Coverage Ratio	50%						60%					
Terrestrial Capacity (Mbps)	100		120		140		100		120		140	
Satellite Altitude (km)	663	677	663	694	714	677	714	694	740	714	867	
Orbit Number	12	6	12	6	18	12	6	6	12	12	6	
Satellite Number/Orbit	18	17	18	16	15	17	15	16	14	15	11	
Total Number of Satellites	Integrated		318		294		270		294		264	
	Satellite-only										420	

Fig. 2. Optimal constellation deployment solutions.

satellites by $W = \max_{l,s} \{ \lceil C_{sate}^{(s)} / C_l \rceil \}$. Using the dynamic programming method, the optimal vector $\mathbf{k}$ can be obtained by calculating $\chi(L, C_{th})$ recursively according to relation (17). Thus, the complexity of the DP-SNM algorithm is $O(SLW^2)$ for the L deployable satellite layers.

V. SIMULATION RESULTS

In this section, we present the optimal satellite deployment scheme for global connectivity and show the minimum number of required satellites satisfying the backhaul rate requirement.

The main parameters and their default values are set as 3GPP R-15 in Table I [11]. For fulfilling the rate requirements of the backhaul services, we consider that $C_{th} = 200$ Mbps for all the UTs. We apply the rain fading parameters given by ITU-R P.618 [10]. We discuss the number of deployed satellites given different terrestrial capacity $C_{terr} \sim (100, 180)$ Mbps and different global average coverage ratio of the terrestrial network $\rho \sim (50\%, 60\%)$ with the pathloss factor $\alpha = 2.5$.

The optimal constellation deployment solutions are presented in Fig. 2 given different terrestrial capacity C_{terr} . It is shown that compared with the satellite-only network, the required number of satellites largely decreases when considering the terrestrial network. As the coverage ratio of the terrestrial network increases, the altitude of satellites rises up and the number of satellites decreases. Fig. 3 presents the optimal deployed altitude and number of satellites on each orbit given different terrestrial capacities with the coverage ratio $\rho = 50\%$. We can observe from the optimal constellation that as the terrestrial capacity increases, the deployed satellite altitude increases so that fewer satellites are required to deploy.

In Fig. 4, we evaluate the required satellite number given different terrestrial capacities in different regions. Assume that $C_{terr}^{low} = 180$ Mbps and $\rho = 60\%$ in the low-latitude and mid-latitude regions (i.e., between $60^\circ S$ and $60^\circ N$). Fig. 4 shows the total number of satellites vs. the terrestrial capacity in high-latitude regions (i.e., larger than 60°) given different high-latitude terrestrial network coverage ratio. It is shown that the satellite number reduces with a decreasing rate as C_{terr}^{high} grows. A maximum high-latitude terrestrial capacity C_o^* exists where the satellite number cannot reduce anymore when

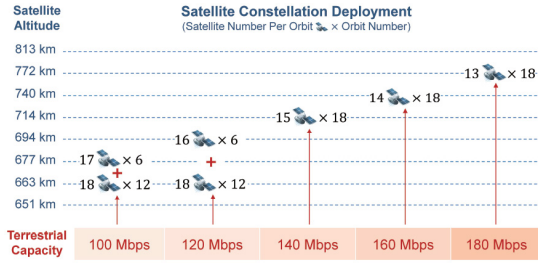


Fig. 3. Satellite altitude of the optimized constellation vs. terrestrial capacity.

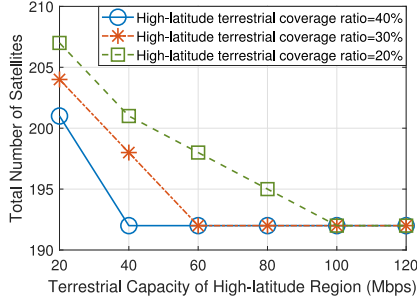


Fig. 4. Total number of satellites vs. high-latitude terrestrial capacity.

$C_{terr}^{high} > C_o^*$, representing that more larger terrestrial capacity in the high-latitude regions cannot contribute to saving satellites anymore. Moreover, with the larger coverage ratio of the terrestrial network, the satellite number reaches the minimum at a smaller high-latitude terrestrial capacity.

VI. CONCLUSION

In this letter, we studied the integrated satellite-terrestrial network. A theoretical framework for the uplink capacity analysis was proposed considering the terrestrial capacity. A mega-constellation with multi-layer satellites was designed given any terrestrial capacity, where the differentiated data rate requirements of all UTs can be satisfied even in the maximum interference case. The following conclusions are drawn. *First*, the optimal constellation deployment is obtained with the minimum number of satellites, where the deployed altitude increases with the terrestrial capacity. *Second*, as the terrestrial infrastructures are distributed more evenly in latitude, the mega-constellation first requires fewer satellites and then remains unchanged regardless of the terrestrial transmission capacity of high-latitude user terminals.

APPENDIX A

Interference From Co-Layer Satellite Links ($n = l$): Let $d_{m,q}^l$ denote the distance between the UT q and satellite m on the l -th layer. The region of co-layer interfering UTs is expressed by $D_{m,q}^l = \{q' \mid d_o^l \leq d_{m,q'}^l \leq \sqrt{h_l^2 + 2Rh_l}\}$, where d_o^l represents the smallest distance from the UT of the neighbouring satellite cell to satellite m , and $\sqrt{h_l^2 + 2Rh_l}$ indicates the longest distance of the interference LoS path.

We then derive the expression of d_o^l by geometry. Given the uniform distribution, let K_l denote the total number of satellites on layer l expressed by $K_l = \frac{k_l}{2\pi(R+h_l)^2(1-\cos\phi_l^{max})}$. $4\pi(R+h_l)^2 = \frac{2k_l}{1-\cos\phi_l^{max}}$ and the distance between the coverage center of any satellite is equidistant to its surrounding coverage centers, expressed by $b_l = \sqrt{\frac{8\pi R^2}{3K_l}}$. The

radius of each satellite's coverage circle is $r_l = R \cdot \phi_l^{max}$. Thus, the radius of the interference-free coverage $r_l^o = \max\{0, b_l - r_l\}$ and the angular radius $\phi_l^o = \frac{r_l^o}{R}$. According to the law of cosines, we can derive the distance $d_o^l = \sqrt{R^2 + (R+h_l)^2 - 2R(R+h_l)\cos\phi_l^o}$.

Interference From Cross-Layer Satellite Links ($n \neq l$): The serving range of satellites on different layers are completely overlapped, so the interfering range is the whole LoS area. Let $D_{m,q}^n$ indicate the region of UTs inducing the cross-layer interference, which is expressed by $D_{m,q}^n = \{q' \mid h_l \leq d_{m,q'}^l \leq \sqrt{h_l^2 + 2Rh_l}\}$, $n \neq l$, where h_l indicates the altitude of the satellite m on layer l , and $\sqrt{h_l^2 + 2Rh_l}$ indicates the longest distance of the LoS path between the interfering UT q' and satellite m .

APPENDIX B

According to the geometry relation, the area of serving satellites on layer l for UT q (in set $\mathcal{A}_q^l$) can be calculated by $S_q^l = 2\pi(R+h_l)^2(1-\cos\phi_l^{max})$. Thus, the average spectrum efficiency $\mu_l^{(s)}$ for layer l can be calculated by $\mu_l^{(s)} = \frac{1}{S_q^l \ln 2} \int_{m \in \mathcal{A}_q^l} \ln(1 + \frac{PG_0(d_{m,q}^l)^{-\alpha}}{K(\sigma^2 + \mathbb{E}[I_{m,q}^l])}) dm \frac{\pi(R+h_l)}{S_q^l R \ln 2} \int_{(h_l)^2}^{(d_m^l)^2} \ln(1 + \frac{PG_0 \cdot v^{-\frac{\alpha}{2}}}{K(\sigma^2 + \mathbb{E}[I_{m,q}^l])}) dv$. Denote the longest distance of between satellite m (on layer l) and its serving UT by $d_m^l = -R \sin \theta_{min} + \sqrt{(R \sin \theta_{min})^2 + h_l^2 + 2Rh_l}$.

By utilizing the integral formula, let $X = \frac{PG_0}{K(\sigma^2 + \mathbb{E}[I_{m,q}^l])}$ and $Y = -\frac{\alpha}{2}$. Define that $\mathcal{F}(v, X, Y) = \int \ln(1 + Xv^Y) dv = v[Y \cdot {}_2F_1(1, \frac{1}{Y}; 1 + \frac{1}{Y}; -Xv^Y) + \ln(1 + Xv^Y) - Y]$. Thus, the average spectrum efficiency $\mu_l = \frac{\mathcal{F}((d_m^l)^2, X, Y) - \mathcal{F}((h_l)^2, X, Y)}{2R(R+h_l)(1-\cos\phi_l^{max}) \ln 2}$.

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